Surge Impedance Loading (SIL)

Class Notes

Notes by BALAJI SIR

Ideal Parameters : $f = 50\ Hz$, $v = 3 \times 10^8\ m/s$

$$\beta' = \frac{2\pi \times 50}{3 \times 10^8}\ l\ rad$$

A = D of lossless T/L

From Nominal π N/W :

$$A = D = \left(1 + \frac{YZ}{2}\right) = 1 + \frac{(j\omega C_1 l)(j\omega L_1 l)}{2} = 1 - \frac{\omega^2 L_1 C_1 l^2}{2}$$

$$\boxed{A = D = 1 - \frac{(\beta')^2}{2}}$$

where β' is in rad

Surge / Lossless / flat / Infinite / Resonant / Tuned / Natural line / Ideal line Impedance (Z_s)

$$Z_c = \sqrt{\frac{Z_1}{Y_1}} = \sqrt{\frac{R_1 + j\omega L_1}{G_1 + j\omega C_1}}$$

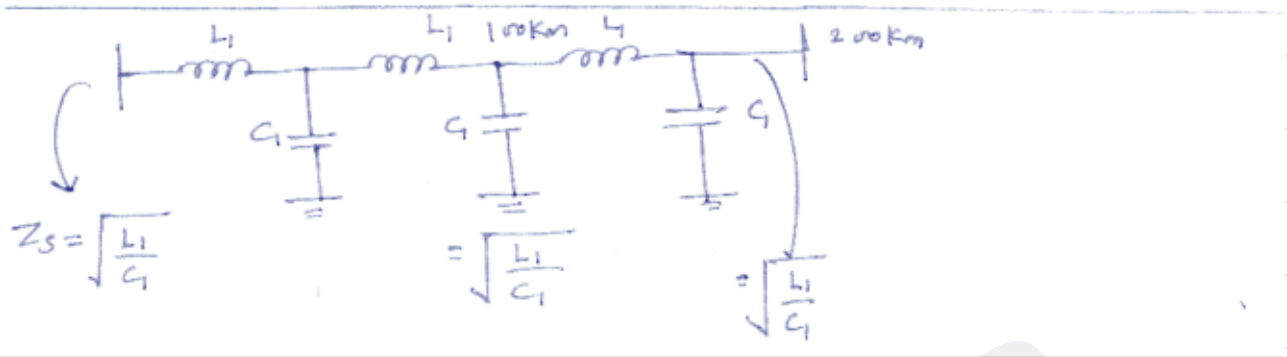
Substitute $R_1 = G_1 = 0$ for lossless T/L

$$\boxed{Z_c = \sqrt{\frac{L_1}{C_1}} = Z_s}$$

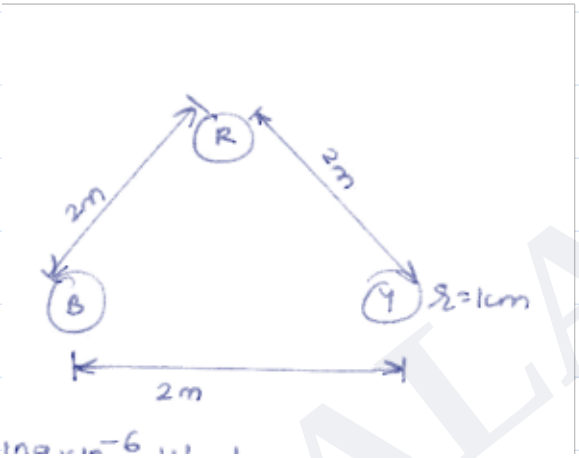
$Z_s = 300 - 400\ \Omega$ for overhead T/L ; $= 40 - 50\ \Omega$ for cable

NOTE : Z_s is independent of length of T/L

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Lossless line of 200 km: L1, C1 sections; the impedance Zs = sqrt(L1/C1) looking in at any point



Equilateral arrangement R, Y, B of 2 m sides, r = 1 cm

$$L_1 = 1.109 \times 10^{-6}\ H/m/ph, \quad C_1 = 10.49 \times 10^{-12}\ F/m/ph$$

$$Z_s = \sqrt{\frac{1.109 \times 10^{-6}}{10.49 \times 10^{-12}}} \approx 325\,\Omega$$

$$v = \frac{1}{\sqrt{1.109 \times 10^{-6} \times 10.49 \times 10^{-12}}} = 2.93 \times 10^{8}\ m/sec$$

* A Tr. line which is independent of frequency, i.e. which allows all the frequency, is called distortionless line.

* It is practically used in communication application (Dish cable).

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* If the series element $(R + jX_L)$ time const. i.e. L/R is equal to the shunt element (C, G) time const. $[T.C = C/G]$, then line becomes independent of freq.

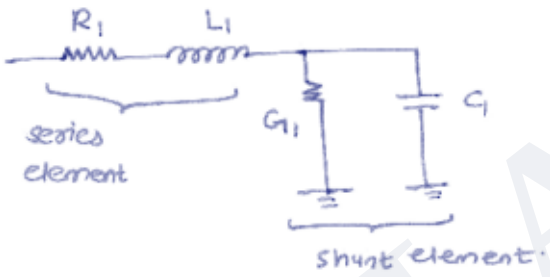
$$Z_c = \sqrt{\frac{R_1 + j\omega L_1}{G_1 + j\omega C_1}} \leftarrow \text{depends on frequency}$$

$$= \sqrt{\frac{R_1}{G_1}} \sqrt{\frac{\left(1 + j\frac{\omega L_1}{R_1}\right)}{\left(1 + j\frac{\omega C_1}{G_1}\right)}}$$

If $\frac{L_1}{R_1} = \frac{C_1}{G_1}$ (condition for distortionless T/L)

$$\boxed{Z_c = \sqrt{\frac{R_1}{G_1}} = Z_D} \leftarrow \text{Independent of frequency}$$

Z_D = Distortionless line Impedance



Series element R1, L1; shunt element G1, C1

4] $L_{circuit} = \frac{\mu_0}{\pi} \ln \frac{d}{r}$, H/m

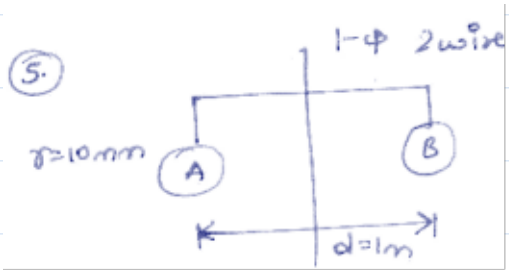
$$= 4 \times 10^{-7} \ln \frac{1.5}{0.7788 \times 0.5 \times 10^{-2}} = 2.38 \times 10^{-6} \text{ H/m}$$

$l = 10$ km

$L = 2.38 \times 10^{-6} \times 10 \times 10^3 = 23.8$ mH

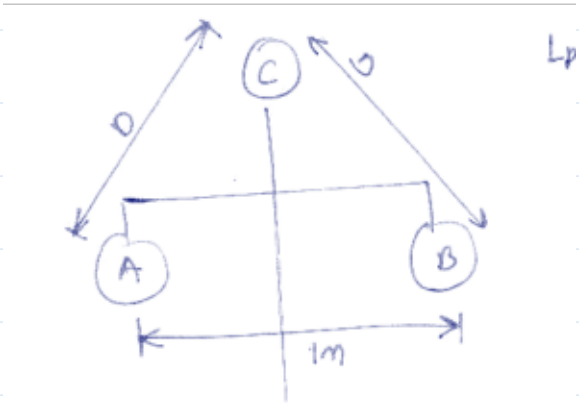
(5) 1-ϕ 2-wire $r = 10$ mm, $d = 1$ m

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Conductors A, B of radius 10 mm, spacing d = 1 m

$$L_A = L_B = 2\times10^{-7}\ln\left(\frac{1}{0.7788\times10\times10^{-3}}\right) = 0.97\times10^{-6}\ H/m$$



Third conductor C added above, A and B 1 m apart, distance D from C

$$L_{ph} = 2\times10^{-7}\ln\frac{(D\cdot D\times1)^{1/3}}{0.7788\times10\times10^{-3}} = 1.05\times L_A = 1.05\times0.97\times10^{-6}\quad (5\%\ \text{increase})$$

$$D = 1.439$$

$$SIL = 3\left(\frac{V_L}{\sqrt{3}}\right)^2\times\frac{1}{Z_s}$$

$$SIL = \frac{V_L^2}{Z_s} = V_L^2\sqrt{\frac{C_1}{L_1}} = V_s^2\sqrt{\frac{C_1}{L_1}}$$

$$SIL \propto V_s^2 \propto \sqrt{C_1} \propto \frac{1}{\sqrt{L_1}}$$

$L_1\downarrow$: Bundle conductor, C_{se}, parallel T/L

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Effect of compensator on SIL :

Z_s = Surge Impedance without compensator = $\sqrt{\frac{L_1}{C_1}}$

Z_{sc} = Surge Impedance with compensator

(SIL) = (SIL) w/o compensator

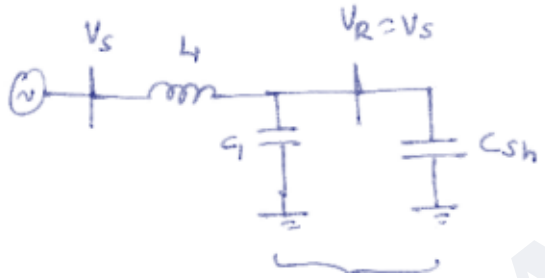
$(SIL)_c$ = SIL with compensator

K_{Csh} = % or Degree of C_{sh} compensator

K_{Lsh} = % or Degree of L_{sh} compensator

K_{se} ... C_{se}

With C_{sh}



Vs - L1 - C1 and Csh at the receiving end (VR = Vs)

$$C_T = C_1 + C_{sh} = C_1\left[1 + \frac{C_{sh}}{C_1}\right] = C_1[1 + K_{Csh}], \quad K_{Csh} = \frac{C_{sh}}{C_1}$$

$$Z_{sc} = \sqrt{\frac{L_1}{C_T}} = \sqrt{\frac{L_1}{C_1(1 + K_{Csh})}}$$

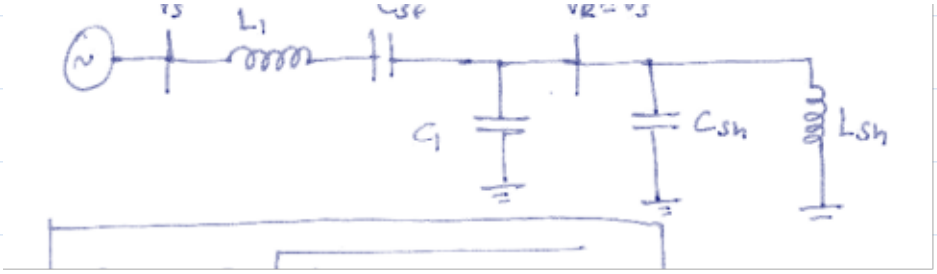
$$\boxed{Z_{sc} = \frac{Z_s}{\sqrt{(1 + K_{Csh})}}}$$

$$(SIL)_c = \frac{V_s^2}{Z_{sc}} = \frac{V_s^2}{Z_s}\sqrt{1 + K_{sh}}$$

$$\boxed{(SIL)_c = (SIL)\sqrt{1 + K_{Csh}}}$$

With C_{sh}, C_{se}, L_{sh}

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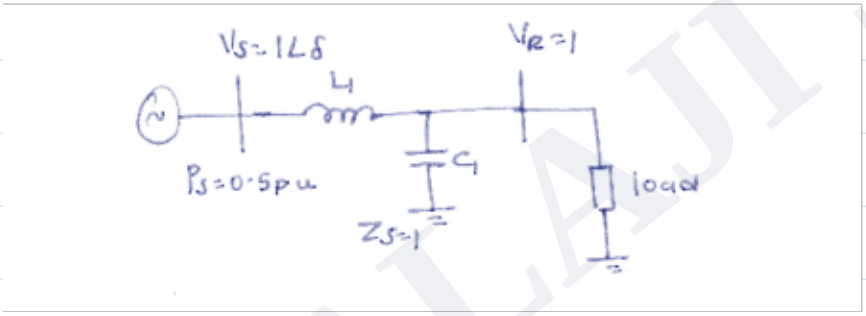


Vs - L1 - Cse - C1 - VR = Vs with Csh and Lsh at the receiving end

$$Z_{sc} = Z_s \sqrt{\frac{(1-K_{se})}{(1+K_{Csh})(1-K_{Lsh})}}$$

$$(SIL)_c = (SIL) \sqrt{\frac{(1+K_{Csh})(1-K_{Lsh})}{(1-K_{se})}}$$

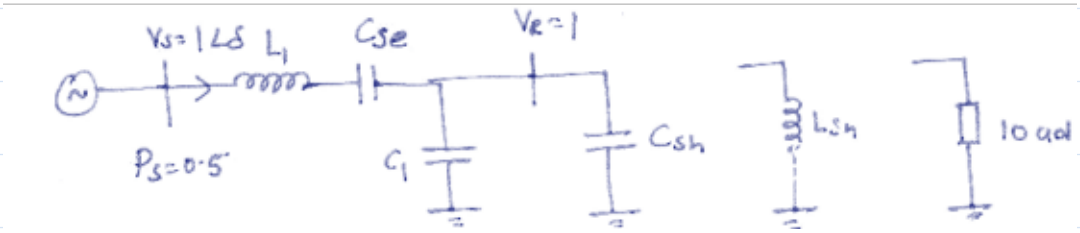
W/o compensator :



Vs = 1 at delta - L1 - C1 - VR = 1 - load; Ps = 0.5 p.u, Zs = 1

$$P_{max} = SIL = \left| \frac{V_s^2}{Z_s} \right| = \frac{1^2}{1} = 1 \text{ p.u}$$

$$P_s = P_{max} \sin\delta \; : \; 0.5 = 1 \times \sin\delta \Rightarrow \delta = 30^\circ$$



Compensated line: Vs = 1 at delta, L1, Cse, C1, VR = 1, Csh, Lsh, load; Ps = 0.5

$$K_{se} = K_{Csh} = K_{Lsh} = 30\% = 0.3$$

Component Z_{sc} $(SIL)_c = P_{max}$ $\delta = \sin^{-1}\left(\frac{P_s}{P_{max}}\right)$

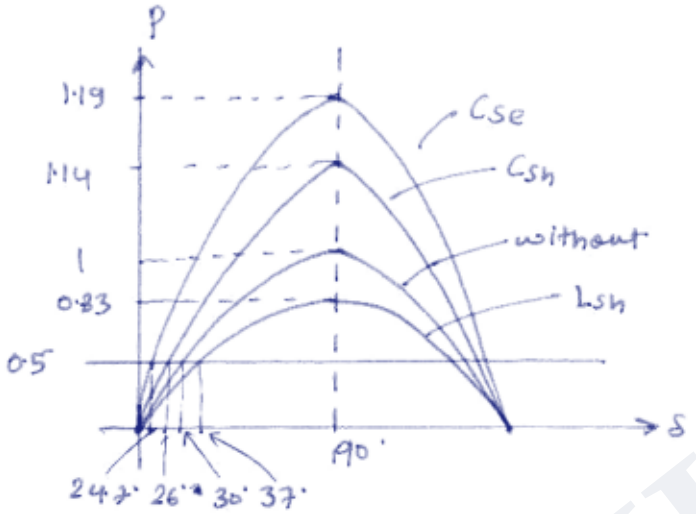
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w/o compensator 1 1 30°

$K_{Csh} = 30\%\ Z_{sc} = \frac{1}{\sqrt{1+0.3}} = 0.87\ 1 \times \sqrt{1+0.3} = 1.14\ \delta = \sin^{-1}\left(\frac{0.5}{1.14}\right) = 26°$

$K_{se} = 30\%\ Z_{sc} = 1 \times \sqrt{1-0.3} = 0.83\ \frac{1}{\sqrt{1-0.3}} = 1.19\ \delta = \sin^{-1}\left(\frac{0.5}{1.19}\right) = 24.7°$

$K_{Lsh} = 30\%\ Z_{sc} = \frac{1}{\sqrt{1-0.3}} = 1.19\ \sqrt{1-0.3} = 0.83\ \delta = \sin^{-1}\left(\frac{0.5}{0.83}\right) = 37°$



P - delta curves: Cse (peak 1.19), Csh (1.14), without compensator (1), Lsh (0.83); Ps = 0.5 line cuts at 24.7, 26, 30 and 37 degrees

Power Transfer capability and stability: $C_{se} > C_{sh} > (w/o) > L_{sh}$

NOTE :

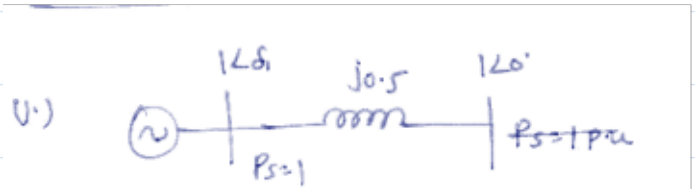
MW loading = SIL => $V_R = V_S$ -> No compensation required

Loading < SIL => $V_R > V_S$ => L_{sh} required for $V_R = V_S$ (off peak load loading time)

Loading > SIL => $V_R < V_S$ => C_{sh} required for $V_R = V_S$ (peak loading time)

Midpoint compensation :

(1)

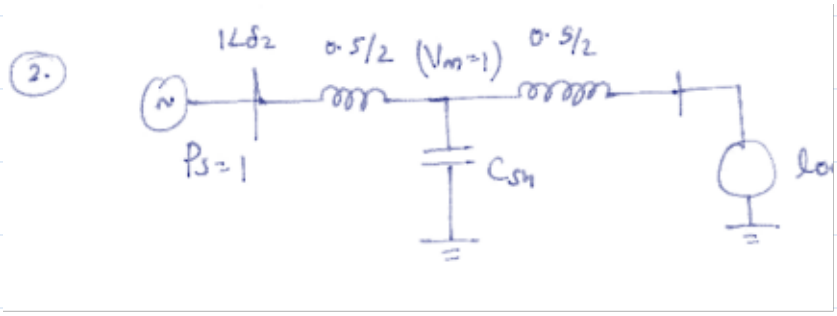


1 at delta1 - j0.5 - 1 at 0 degrees; Ps = 1 p.u

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$$P_s = \frac{1 \times 1}{0.5} \sin(\delta_1) \Rightarrow \delta_1 = 30^\circ$$

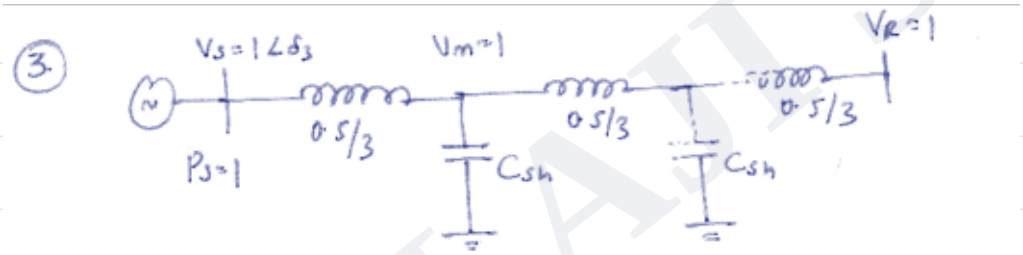
(2)



1 at delta2 - 0.5/2 - (Vm = 1, Csh) - 0.5/2 - load; Ps = 1

$$P_s = 1 = \left|\frac{1 \times 1}{0.5/2}\right| \sin(\delta_2) \quad \delta_2 = 14.7^\circ \approx 15^\circ \Rightarrow \frac{\delta_1}{2}$$

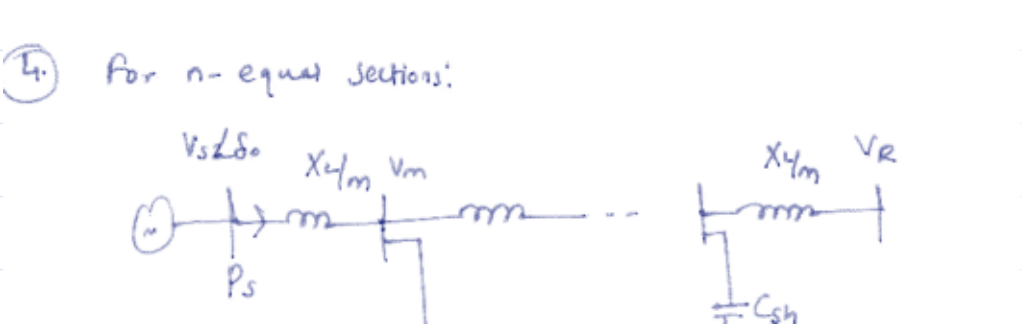
(3)



Vs = 1 at delta3 - 0.5/3 - Vm = 1, Csh - 0.5/3 - Csh - 0.5/3 - VR = 1

$$P_s = 1 = \left|\frac{1 \times 1}{0.5/3}\right| \sin\delta_3 \quad \delta_3 = 9.8^\circ \approx 10^\circ = \left(\frac{\delta_1}{3}\right)$$

(4) For n-equal sections :



Vs at delta0 - XL/m - Vm, Csh - XL/m ... - Csh - XL/m - VR

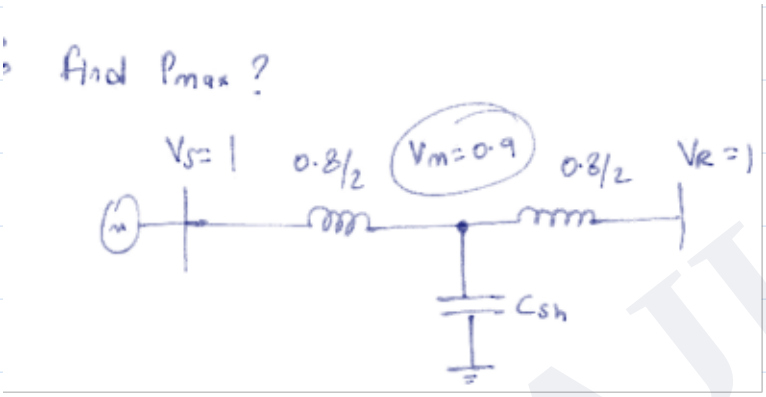
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$$P_s = \left|\frac{V_s V_m}{X_L/m}\right| \sin(\delta_0) \qquad V_s = V_m$$

$$\boxed{P_s = \left|\frac{m V_s V_m}{X_L}\right| \sin\left(\frac{\delta}{m}\right)}$$

$$P_s = \left|\frac{m V_s^2}{X_L}\right| \sin\left(\frac{\delta}{m}\right)$$

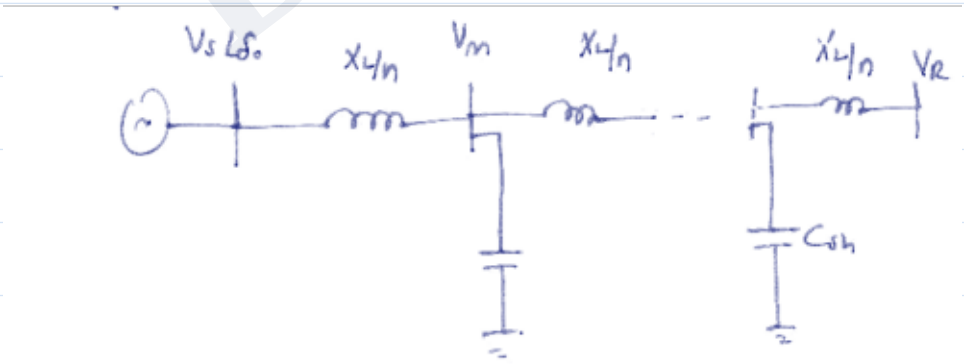
Eg: Find P_{max} ?



Vs = 1 - 0.8/2 - (Vm = 0.9, Csh) - 0.8/2 - VR = 1

$$P_{max} = \left|\frac{V_s V_m}{X_L}\right| = \left|\frac{1 \times 0.9}{0.8/2}\right| = \frac{9}{4} = 2.25 \text{ p.u}$$

For n equal sections :



Vs at delta0 - XL/n - Vm - XL/n ... - XL/n - VR, with Csh at each section

$$P_s = \left|\frac{V_s V_m}{X_L/n}\right| \sin(\delta_0)$$

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